GLOBAL RENEWABLE ENERGY IN 2025: HOW FAR HAVE WE COME?

A data-driven overview of renewable energy's growing share in the global energy and electricity mix, spanning hydropower, wind, and solar technologies.

Source: Our World in Data – Renewable Energy

75%

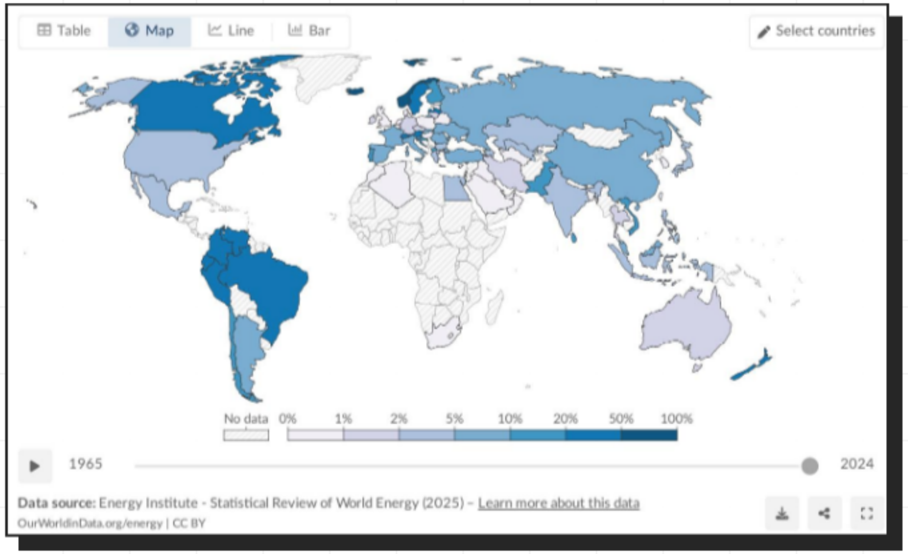
OF GLOBAL GHG EMISSIONS STILL COME FROM FOSSIL FUELS

- **THE DOMINANCE:** Fossil fuels have dominated the energy system since the Industrial Revolution.
- **THE CONSEQUENCE:** Burning fossil fuels drives climate change and causes at least 5 million premature deaths annually from air pollution.
- **THE SOLUTION:** The world must rapidly shift toward low-carbon energy sources—nuclear and renewable technologies.

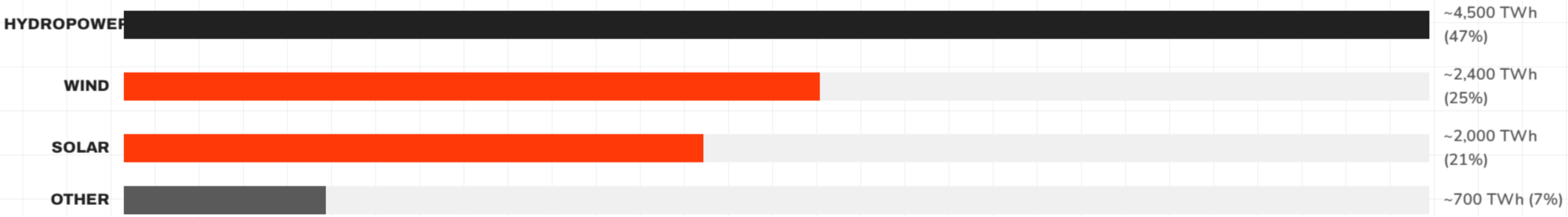
RENEWABLES NOW SUPPLY ~14% OF GLOBAL PRIMARY ENERGY

Approximately one-seventh (~14%) of the world's primary energy now comes from renewable technologies (2024 data, substitution method).

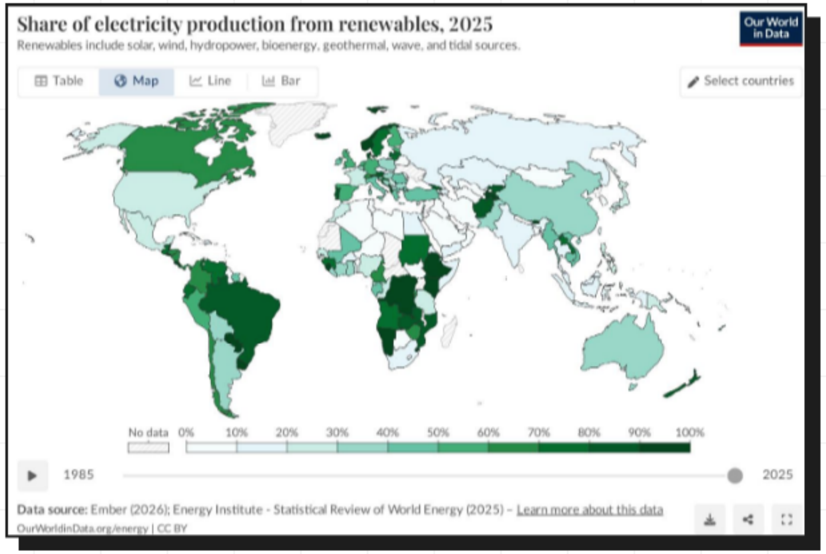
Primary energy encompasses electricity, transport, and heating combined. Because transport and heating are harder to decarbonize (more reliant on oil and gas), the renewable share in the total energy mix is lower than in the electricity mix alone.



RENEWABLE ELECTRICITY GENERATION IS GROWING RAPIDLY, LED BY WIND AND SOLAR



Wind and solar combined now nearly match hydropower's output, accelerating dramatically since 2010.



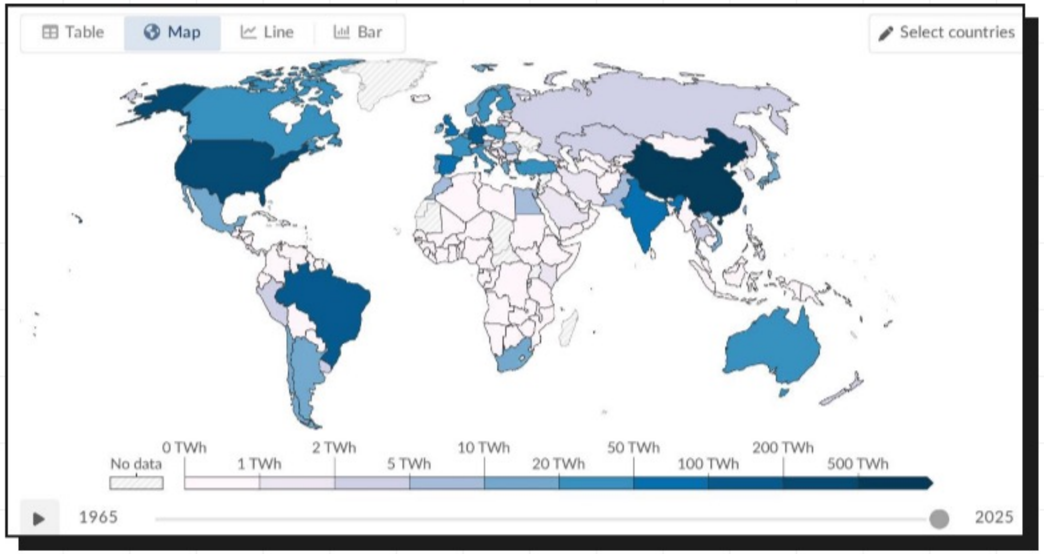
ALMOST ONE-THIRD OF GLOBAL ELECTRICITY NOW COMES FROM RENEWABLES

Electricity is the sector where renewables have made the most progress. Regional leaders exceed 60–90%+, while laggards remain below 10%.

REGION/COUNTRY	RENEWABLE SHARE
Norway	~98%
Brazil	~85%
Canada	~65%
China	~35%
India	~25%
Saudi Arabia	<2%

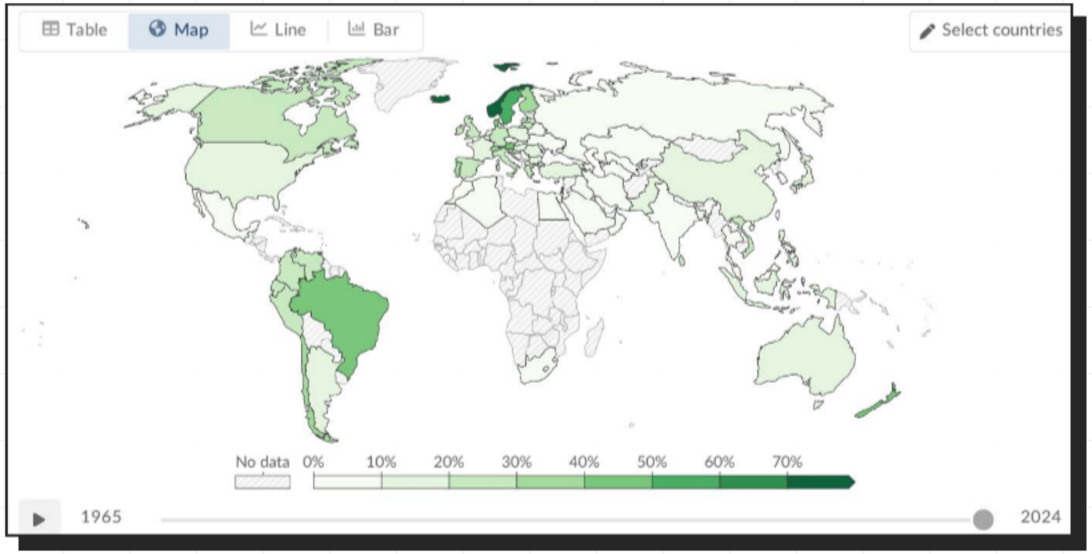
HYDROPOWER REMAINS THE LARGEST RENEWABLE SOURCE AT ~50% OF GENERATION

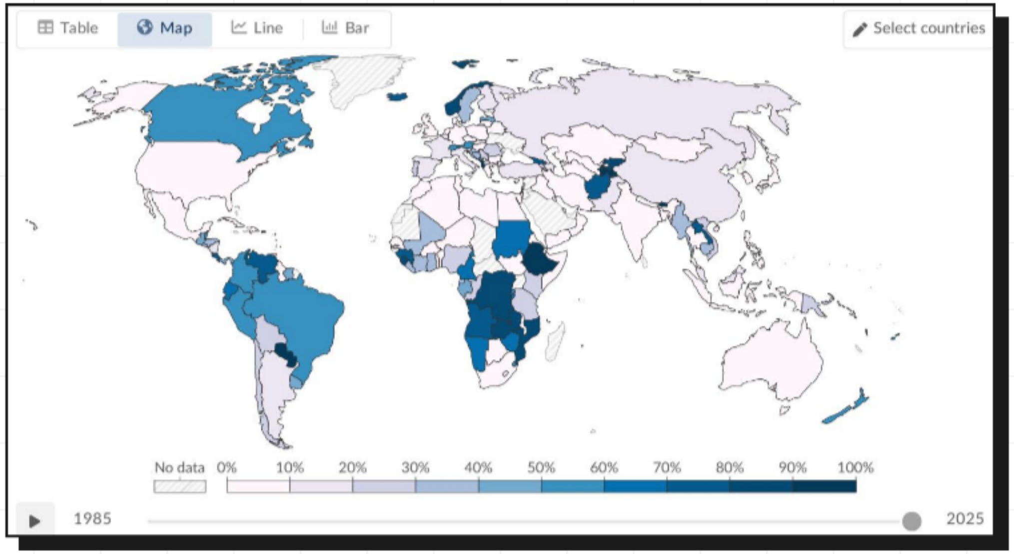
- **DOMINANT FORCE:** Hydropower accounts for roughly half of all renewable electricity globally.
- **TOP PRODUCERS:** China, Brazil, and Canada lead the world in hydropower generation.
- **LEGACY:** Generating electricity at scale for over a century, though growth potential is now outpaced by other technologies.



WIND AND SOLAR ARE THE FASTEST-GROWING ENERGY TECHNOLOGIES IN HISTORY

- **WIND POWER:** Generates ~2,400 TWh (~25% of renewable electricity), with massive offshore and onshore expansion.
- **SOLAR POWER:** Generates ~2,000 TWh (~21%). Solar is the fastest-growing source worldwide.
- **COMBINED IMPACT:** Together, they now nearly match hydropower output.
- **KEY DRIVERS:** Dramatic growth since 2010 fueled by rapidly falling costs and strong policy support.





STARK REGIONAL GAPS: SUB-SAHARAN AFRICA LAGS WHILE SCANDINAVIA AND LATIN AMERICA LEAD

- **HYDROPOWER HUBS:** Scandinavian countries and Brazil benefit from abundant hydro, exceeding 60–90% renewables.
- **WIND & SOLAR PIONEERS:** European nations like Denmark and Germany have heavily invested to transform their mix.
- **THE INFRASTRUCTURE GAP:** Sub-Saharan Africa has minimal renewable infrastructure. The Middle East remains almost entirely fossil-fuel dependent (<10%).

KEY TAKEAWAYS: RENEWABLES ARE GROWING FAST BUT THE TRANSITION MUST ACCELERATE

1. MAJOR MILESTONE

Renewables supply ~14% of primary energy and ~30% of electricity globally.

2. CHANGING MIX

Hydro remains dominant (~50%), but wind and solar are rapidly catching up.

3. REGIONAL DIVIDE

Scandinavia and Latin America lead; Middle East and Sub-Saharan Africa lag.

4. THE GAP

Transport and heating lag far behind electricity in the renewable transition.

THE PACE MUST ACCELERATE SIGNIFICANTLY TO MEET CLIMATE TARGETS.